

Hardware opportunities

Design and implementation: designing telecom systems equipment, infrastructure, and networks is a good career option since there has to be a proper design in place before installation of equipment such as switchers and routers at a base station. But design and manufacturing of original telecom equipment is yet to take off on a large scale in India. Equipment design engineers, therefore, see better prospects abroad.

❑ **Operations and maintenance:** the telecom infrastructure requires operational support and maintenance to function at optimal capacity. This domain includes monitoring, troubleshooting, and analysis of the operations which take place at the base service station. Ultimately, the job of those in the operation and maintenance department is to ensure that voice is transmitted from one end to another.

❑ **Services:** increased competition means telecom service providers must provide quality service keeping operational costs under control. They deploy solutions for service quality management, fault management, and performance management.

❑ **Network:** job functions in wired, wireless, and data networks all fall under the network umbrella. Employed in this domain are engineers who specialise in wireless and other technologies. They are, for example, GSM or CDMA engineers / specialists, GPRS engineers / specialists. Specifically, installation, planning, administration, security management, troubleshooting, and testing of networks are involved in this domain.

Freshers are usually absorbed into the industry in networking jobs. Network engineers at times design terrestrial networks and certify them. Vendor-specific certifications are a plus in a network engineer's profile. Of all the network functions, security is an area worthy of mention in terms of opportunities.

Face-2-Face with Practitioner

"I specialise in delivering software for telecommunication products. A significant percentage of people in the telecommunication industry fall in this category..."

Abhinav Goyal
Senior Technical lead
Sonous Networks India

Software opportunities

Software in this industry usually enables signal testing and network monitoring. Real-time operating systems are of particular importance to aspirants.

Revenue and billing services: those skilled in DBA (Database Analysis) software systems and data warehousing are employed extensively in this sector. They specialise in working on vendor-based applications and software systems such as Oracle, Kenan, and Siebel. The professional here keeps a check on whether customers are correctly billed and prevents fraud or faulty billing.

Value-Added Services (VAS): a fiercely-growing domain, emerging as a separate industry, VAS includes entertainment, commerce, TV, and more — on the mobile. The majority of the opportunities in VAS are towards programming for the mobile platform.

Facts and figures

Total telephony subscribers:

(Wireless and Landline) were 400.05 million (Jan 2009)

Cellphones: 362.3 million (Jan 2009)

Land Lines: 37.75 million (Jan 2009)

Broadband connection: 6.22 million (March 2009).